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Creating Large Protein Libraries: An Efficient Engineering Approach

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Abstract

Proteins are vital to all biological processes, yet naturally occurring proteins may not always be optimal for specific purposes. Mutant libraries, containing various protein variants, are increasingly used to optimize protein functions. Directed evolution is a common approach to evolve proteins. Random mutagenesis methods, such as UV exposure, chemical mutagenesis, or error-prone PCR, generate diverse libraries without needing prior information about the structure or function of the protein, but they often produce many undesirable mutants. Targeted methods yield specific mutations with high accuracy. SLUPT (Synthesis of Libraries via deoxyuridine-containing PCR Templates) is a recent method that efficiently creates multi-site mutations without relying on any specific bacterial strains or restriction sites. This method involves creating single-stranded DNA using dU-containing bases, annealing mutagenic primers, filling gaps and ligating the ends. We aim to establish an efficient protocol using SLUPT for generating large protein libraries, validated by creating GFP mutants with varied fluorescence properties. These mutagenesis approaches will be essential for future work generating large libraries suitable for characterization by robotic systems in high-throughput, allowing exploration of complex genetic fitness landscapes.

Introduction

Proteins are essential for all biological processes. However, naturally occurring proteins and other biomolecules are not always optimal for specific purposes and may require additional functionality. Mutant libraries, containing various protein variants, are increasingly being constructed to optimize protein functions. Protein engineering involves screening these mutant libraries to discover novel proteins with enhanced expression levels and additional functionality, such as solubility, stability, or improved enzymatic activity. Directed evolution is often used to evolve and select proteins for desired phenotypes. Advances in protein engineering have enabled us to study point mutations and perform deep mutational scanning of millions of variants.¹⁻³

Achieving these goals often requires modifying existing proteins to create mutant variants with potentially improved properties. However, the vast number of possible mutations presents a significant challenge. Approaches for producing these libraries can be either targeted or random. Random approaches, such as UV and chemical mutagenesis, or error-prone PCR using low-fidelity polymerases, are often employed. DNA shuffling, which creates libraries through the random fragmentation of DNA of homologous sequences, is also used. These generic methods do not require prior structural or functional information to create the library. However, mutation of two consecutive bases is rare, making it difficult to generate all possible amino acid combinations. Additionally, these methods can introduce stop codons and often fail to mutate the desired bases, resulting in the generation of many nonfunctional or undesirable mutants.¹⁻³

Targeted approaches are often used in cases where the structure, structure-function relationship, or important regions affecting the desired property in DNA are known. These approaches are highly accurate and can be easily controlled to produce any desired sequence. Synthetic DNA libraries can also be ordered due to the low cost of DNA synthesis; however, they are limited by the size of the DNA and the mutations that can be introduced.^{4,5} Targeted mutagenesis can be achieved through PCR with mutated codons. Computational methods⁶⁻⁹ can design these libraries, and the use of primers with degenerate codons ensures that all possible or at least the desired amino acid substitutions are made. Overlap extension PCR can generate the full DNA sequence containing the desired mutations, and custom oligos can be synthesized to produce the required DNA. Most of these methods rely on some

form of gene assembly where a series of overlapping fragments are assembled to form the final product. However, when introducing variations in multiple regions in the DNA, these methods either require multiple PCR cycles to generate intermediate products that are later assembled or a single reaction with a collection of primers that spans the entire gene. This can become tedious, complex, and time-consuming. The intermediate steps also make it difficult to control the precise distribution of randomized nucleotides at the mutated sites, resulting in some sequences being completely absent or an oversampling of certain sequences.^{3-5,10,11}

Other methods to generate DNA libraries with targeted multi-site mutations involve generating single-stranded DNA (ssDNA) and then annealing mutagenic primers, followed by extension and ligation to produce the mutant library. One such method, Quikchange Multi Site-Directed Mutagenesis¹² developed by Invitrogen, generates ssDNA by melting and then annealing the mutagenic primers. However, the parent strand/non-template plasmid DNA can compete with primers during annealing. Another approach by Caucheteur et al.¹³ relies on generating ssDNA using M13 phage and the *dut ung* *Escherichia coli* strain. Nicking mutagenesis^{14,15} is based on similar principles, though the generation of ssDNA depends on the presence of a 7-bp BbvCI restriction site, which is not always present.¹⁶⁻¹⁸ SLUPT (Synthesis of Libraries via deoxyuridine (dU)-containing PCR Templates) can efficiently create multiple, simultaneous, specific substitutions within a target sequence.¹⁹ Unlike other methods, the generation of ssDNA in SLUPT is not dependent on any strain of bacteria, virus, or restriction site but is based on PCR, making it applicable to any DNA sequence. First, the target DNA is amplified using dUTPs and 5' phosphorylated primers. The top strand is digested with lambda exonuclease, which recognizes and digests the top 5' phosphorylated strand. Mutagenic primers are then annealed, gaps are filled with DNA polymerase, and nicks are sealed. The U-containing strand is digested with UDG, resulting in ssDNA with desired mutations. Multiple DNA regions can be altered in one reaction by using one primer for each region requiring modification. The parental sequence is almost completely absent due to the digestion of parent strands. Hence, SLUPT is ideally suited for protein engineering, ensuring that all sequences within the library are at the same concentration, thus maximizing the number of protein variants that can be effectively screened. The SLUPT strategy is illustrated in Figure 1.

The goal of this work is to develop an effective protocol for generating large protein libraries that incorporate targeted mutations. Expansion can be done by focusing on specific sites believed to be crucial while also introducing random variations to avoid biases based solely on prior knowledge. We applied the SLUPT method to create a protein library and established our protocol using GFP as a model. If successful, this approach should yield a diverse array of mutants exhibiting different fluorescent colors, as evidenced by shifts in their emission spectra.

Method Details

Preparation of single stranded dU-containing template

Synthesis and purification of the dU-containing template: All primers used in this study are listed in Table 1. A plasmid containing the GFP gene under a constitutive promoter pSS-01-023, was sourced from the lab. The GFP gene was amplified using dUTP mixture (Thermo Scientific, R0251) and Taq DNA Polymerase (New England Biolabs, M0273S). The amplification primers used were PEGFP1 (forward) and PEGFP2 (reverse), with the forward primer being 5' phosphorylated and the reverse primer unphosphorylated to facilitate digestion by lambda exonuclease. The reaction mixture components are detailed in Table 2, and the PCR conditions are outlined in Table 3. The amplified product was verified on a 1% agarose gel and purified using the GeneJet PCR Purification Kit (Thermo Scientific,

K0701). To scale up the amplification, ten 50 µl PCR reactions were set up using the purified amplified DNA as a template, followed by additional PCR purification.

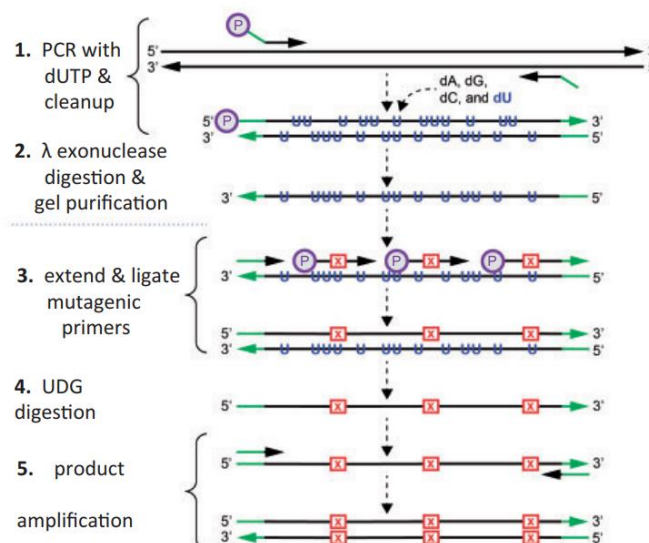


Fig 1. SLUPT strategy: 1) The target DNA is amplified using dUTPs and 5' phosphorylated primers. 2) The top strand is digested with lambda exonuclease, which recognizes and digests the top 5' phosphorylated strand. 3) Mutagenic primers are then annealed, gaps are filled with DNA polymerase, and nicks are sealed. 4) The U-containing strand is digested with UDG, resulting in ssDNA with desired mutations. 5) The obtained ssDNA can be amplified using standard or error-prone PCR

Table 1: Primers used in this study

Primer	Sequence	5' phosphorylation	Purpose
PEGFP1	5' GTTTAATACTAGAGAAAGAGGGGAAATACTAGTACC 3'	yes	End Primer
PEGFP2	5' GAAGGGAGGCCTTATTGTTCTGCTC 3'	No	End Primer
PDGFPC hMF	5' CCAACACTTGTCACACTTT MRSTYRB GGTGTTCATGCT TTGCGAG 3'	Yes	Mutagenic donor primer for the Chromophore region
PDGFPC 03MF	5' GACAACCATTACCTGTCC WMY CAATCTGCCCTTTCGAAAG 3'	Yes	Mutagenic donor primer for the 203 rd amino acid
PDGFPC hCF	5' CCAACACTTGTCACACTTTTCGGTTATGGTGTTCATGCTT TGCGAG 3'	Yes	Non-mutagenic donor primer for the Chromophore region, control
PDGFPC 03CF	5' GACAACCATTACCTGTCCACACAATCTGCCCTTTCGAAAG 3'	Yes	Non-mutagenic donor primer for the 203 rd amino acid, control

Table 2: UTP PCR reaction:

Component	Quantity (50 μ l)	Final Concentration
10x standard taq Buffer	5 μ l	1x
10mM dUTP	1.25 μ l	0.25mM
PEGFP1	5 μ l	0.2 μ M
PEGFP2	5 μ l	0.2 μ M
Template DNA (plasmid)	0.5 μ l	10 ng
Taq Polymerase	0.2 μ l	1U
Nuclease free water	33.05 μ l	

Table 3: UTP PCR condition:

Step	Temp	Time
Initial Denaturation	95 °C	30 seconds
Amplification- 40x	Denaturation- 95 °C Annealing- 54 °C (NEB Ta) Extension- 68 °C	30 seconds 30 seconds 1 min
Final Extension	68 °C	10 mins
Hold	4 °C	infinity

Digestion of the 5'phosphorylated 'top strand' with Lambda exonuclease: 2 μ g of purified dU-containing double-stranded DNA (dsDNA) was digested with Lambda Exonuclease (Thermo Scientific, EN0561) at 37°C. The enzyme concentration and incubation times were varied to optimize the digestion conditions. The reaction mixture components are detailed in Table 4. Post-digestion, the reaction products were analyzed on a 0.7% agarose gel stained with SYBR™ Green II RNA Gel Stain (Thermo Scientific, S7564). The desired ssDNA band was isolated and extracted using the Monarch DNA Gel Extraction Kit (NEB, T1020S). From four digestion reactions, approximately 15 ng/ μ l of dU-containing ssDNA was obtained. It is recommended to assess the quality of the ssDNA prior to proceeding with the generation of the mutant library.

Table 4: Lambda exonuclease digestion reaction

Component	Quantity (15 μ l)	Final Concentration
10x Lambda exonuclease Buffer	1.5	1x
dsDNA	10 μ l	2 μ g
10U/ μ l Lambda exonuclease	0.5-1 μ l	5-10U
Nuclease free water	Upto 15ul	

Testing the quality of ssDNA (optional): To assess the quality and quantity of the obtained single-stranded DNA (ssDNA) for mutant library construction, serial dilutions (1, 1:10, 1:100, 1:1000) of the ssDNA were prepared. Each dilution was either treated with Antarctic Thermolabile Uracil DNA Glycosylase (UDG) (New England Biolabs, M0372S) or left untreated. The treatments were incubated at 37°C for 30 minutes, with the reaction mixture details provided in Table 5. Subsequently, standard PCR amplification was performed on each sample using 1 μ l of the treated ssDNA as the template.

During the initial cycles of PCR, only the complementary primer was added. The PCR program was paused after a few cycles to allow the addition of the second primer, and then continued. The presence of PCR product in non-treated samples would indicate that the ssDNA was of adequate quality for use in library construction. Conversely, the absence of PCR product in UDG-treated samples would confirm the effective removal of uracil-containing DNA, indicating no contamination of the ssDNA template. This step is crucial to determine the appropriate ssDNA dilution for subsequent library generation steps.

Table 5: UDG Test:

Component	UDG digestion (10 μ l)	Non-Digestion (10 μ l)
10x standard taq Buffer	1 μ l	1 μ l
BSA 1mg/ml	1 μ l	1 μ l
ssDNA template (each dilution)	1 μ l	1 μ l
UDG	0.5 μ l	-
Nuclease free water	6.5 μ l	7 μ l

Mutant library generation

Design of donor primer: The donor primers should be designed such that their annealing temperature (excluding the mutated region) is above 55°C, and they should contain 15–20 bases on each side of the desired mutated region that are complementary to the template sequence. All donor primers were ordered with a 5' phosphate, as this is needed for the ligation step. Lyophilized donor primers were resuspended in 10 mM Tris pH 8.5, typically at a 100 μ M concentration, and stored at –20°C.

Previous studies have shown the important regions in GFP which are responsible for the color and fluorescence of the fluorescent protein. The chromophore region (S65–Y66–G67) of GFP is responsible for its fluorescence due to its unique chemical structure and environment within the protein, which allows it to absorb light and then emit it at a different wavelength. Residues that are in the vicinity of the chromophore region and interact with amino acid residues in this region have also been shown to drive the fluorescence property. Residue F64 has been shown to increase the folding of the GFP protein at 37°C, thus enhancing the fluorescence. Mutation of Y66H results in blue fluorescence; F64L, S65T, and Y66W result in cyan fluorescence; and F64L, S65T or S65G, and Y66W result in the yellow variant. Thus, residues selected for mutagenesis were 64, 65, 66, and 203. The degenerate mutagenic donor primers used in this study for the 64-66 region and the 203rd region are shown in Table 1^{20–23}.

Annealing, extension and ligation of the primers: Annealing of donor primers to the ssDNA template was done at 18°C for 30 minutes to favor random annealing of the primer mixtures to the template. Typically, donor primer: ssDNA ratios around 1000:1 work well, but as the primer: template ratio decreases, there is an increased likelihood of skipping a primer and obtaining the template sequence instead of the desired variants. The amount of ssDNA template used here depends in part on the previous UDG test. Typically, ssDNA template at a concentration of ~2.5 ng/ μ l or ~10 fmol/ μ l is used. The reaction mixture is shown in Table 6. For the extension and ligation reaction, in a PCR tube, place 1 μ l of the annealed sample, 10x Taq DNA ligase buffer, dNTP mixture, Taq DNA ligase (New England Biolabs, M0208S), Phusion-U-Hotstart polymerase (Thermo scientific, F555S). Incubate at 55°C for 30'. Reaction mixture is shown in table 7.

Table 6: Annealing reaction

Component	Quantity (10 μ l)	Final Concentration
10x Taq DNA ligase buffer	1 μ l	1x
Forward Primer	1 μ l	1 μ l
ssDNA template (10 fmol/ μ l)	1 μ l	~2.5 ng
5' phosp Donor primer mix	1 μ l each	10pmol/ μ l
Nuclease free water	Upto 10 μ l	

UDG digestion: each gap filled reaction should be digested with UDG for 30' at 37°C to remove the U containing strand and prepare the ssDNA containing desired mutations. Reaction mixture is shown in table 8. ssDNA with desired mutations is ready and now it can either be amplified using standard PCR or error prone PCR to get random mutations as well.

Table 7: Extension and Ligation reaction

Component	Quantity (10 μ l)	Final Concentration
10x Taq DNA ligase buffer	1 μ l	1x
10mM pure dNTPs	0.1 μ l	100 μ M
Phusion U hotstart	0.6 μ l	0.75U
Taq DNA ligase (dilute fresh from 40U/ μ l to 2.5U/ μ l in Taq DNA ligase buffer)	1	2.5U
Annealed sample	1 μ l	
Nuclease free water	Upto 10 μ l	

Table 8: UDG digestion

Component	UDG digestion (10 μ l)
10x standard taq Buffer	1 μ l
BSA 1mg/ml	1 μ l
Gap filled template	2.5 μ l
UDG	0.5 μ l
Nuclease free water	5 μ l

Amplification of single stranded mutant DNA

Standard PCR or error-prone PCR (to increase diversity) can be performed to amplify the mutagenic library. Due to time constraints, only standard PCR was carried out. A 2.5 μ l aliquot of the digested reaction was used as the template for PCR. During the amplification of the PCR product using ssDNA, the complementary primer was added for the initial few cycles. The program was then paused to add the other primer before continuing with the PCR. To increase the yield, the reaction volume was scaled up to 100 μ l, followed by PCR purification.

Cloning of mutant library

The vector backbone was prepared using digestion with SpeI and StuI (NEB: R3133S, R0187S) followed by Antarctic phosphatase treatment. The desired band for vector backbone was gel extracted. The amplified mutant library was digested with SpeI and StuI as well and then gel extracted. The insert was ligated to vector (5:1 molar concentration) using T4 DNA ligase (Thermo scientific, EL0011) at 22°C for

1 hr. The reaction mixture was transformed into chemically competent *E. coli* Dh5 α and the transformed cells were plated on LA plate containing 100 μ g/mL Ampicillin. Plasmid mixture from 10 colonies were also sent for whole plasmid sequencing to determine if desired site directed mutagenesis has worked or not.

Emission Spectrum analysis for cells

The transformed cells on LA plate were analyzed under blue light. 91 Glowing colonies were selected along with controls (Dh5 α , Dh5 α transformed with pSS-01-023 [referred to as GFP], Dh5 α transformed with insert without mutation [referred to as pos control] and inoculated in a 96 well plate containing LB and 100 μ g/mL Ampicillin (if required). Cells were grown for 24 hours and were then pelleted, washed with PBS and then resuspended in minimal M9 media for fluorescence measurement. Emission spectrum was measured at 405nm and 370 nm (for selected colonies only). Shift in peak emission was determined for every colony selected.

Results

GFP library generation

Plasmid pSS-01-023 was used as a template to amplify dU-containing double-stranded GFP DNA (782 bp), with the gel image shown in Fig. 2A. The dU-containing dsDNA was further digested to remove the top strand and produce dU-containing ssDNA, as shown in the gel image in Fig. 2B. Optimal digestion was achieved using 0.5 μ l of enzyme with 30 minutes of incubation. Following gel extraction, 15 ng/ μ l of ssDNA was obtained. The quality of the ssDNA at various dilutions was then assessed with UDG-digested and non-digested templates, followed by PCR to determine the purity and minimum quantity of ssDNA required for producing the mutant library; the gel image is shown in Fig. 2C. All UDG-digested templates gave a negative PCR result, suggesting that the template consisted of only U-containing ssDNA. PCR amplification was observed at a 1:100 dilution of the non-digested template. Annealing of donor primers, extension, ligation, and UDG digestion, as previously described, were performed to produce the site-directed mutant library. The standard amplification of the SDM library is shown in Fig. 2D. The digestion of the amplified SDM library (745 bp) and the vector backbone, followed by Antarctic phosphatase treatment (2266 bp), is depicted in Fig. 2E.

The insert and vector were transformed into Dh5 α following ligation. The transformed colonies, observed under blue light, are shown in Fig. 3. Since most colonies on the control plate were glowing, it indicates that the cloning process was successful. However, the presence of non-glowing colonies in the mutant library could be due to several factors. These may include the introduction of stop codons in the chromophore region of the YRB degenerate codon or a combination of codons/amino acids that results in the loss of fluorescence.

A mixture of 10 glowing and 10 non-glowing colonies was sent for whole plasmid sequencing using Oxford Nanopore technology. The region where site-directed mutagenesis (SDM) was expected to occur is highlighted in Fig. 4. Sequencing results showed multiple peaks at the same position, indicating that the mutagenesis was successful. Interestingly, the sequence coverage plot revealed a ~50 bp deletion in approximately 10% of the plasmid samples from the glowing colonies, occurring in the GFP insert region (Fig. 4C), but not within the targeted mutagenesis region. This suggests that fluorescence could also be achieved with a truncated protein sequence. For the plasmid samples from non-glowing cells, the sequence coverage plot (Fig. 4D) indicated several issues: some colonies had self-ligated

vector backbones (missing the ~745 bp insert), some had a ~50 bp deletion at a similar position as seen in the glowing colonies, and others had a ~600 bp deletion.

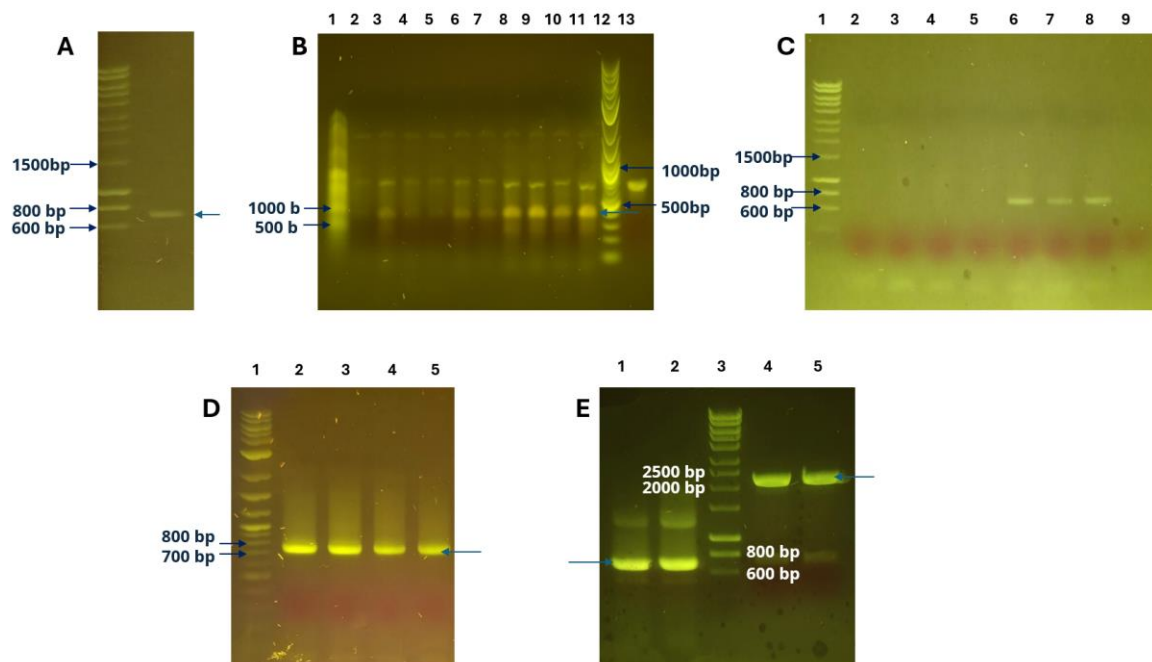


Fig 2: A) Amplification of dU containing dsDNA. B) Lambda exonuclease digestion of dsDNA at different enzyme concentration and incubation time (1: ssRNA ladder, 2: 4hr 0.5 μ l, 3: 2 hr 0.5ul, 4: 2h 0.75ul, 5: 2h 1ul, 6: 1.5h 0.75ul, 7: 1.5h 1ul, 8: 1.5h 0.5ul, 9: 30min 0.75, 10: 30min 1, 11: 30min 0.5ul, 12: DNA ladder, 13: dsDNA). C) quality check ssDNA (1: DNA ladder, 2: UDG digested 0 dilution, 3: UDG digested 1:10 dilution, 4: UDG digested 1:100 dilution, 5: UDG digested 1:1000 dilution, 6: non-UDG digested 0 dilution, 7: non UDG digested 1:10 dilution, 8: non UDG digested 1:100 dilution, 9: non UDG digested 1:1000 dilution). D) Amplification of library (1: DNA ladder, 2, 3: Control library without mutant donor primer, 4,5: mutant library with mutant donor primer). E) digestion of insert and vector with *Stu*I and *Spe*I (1: insert digestion of non-mutant control library, 2: insert digestion of mutant library, 3: DNA ladder, 4, 5: vector backbone)

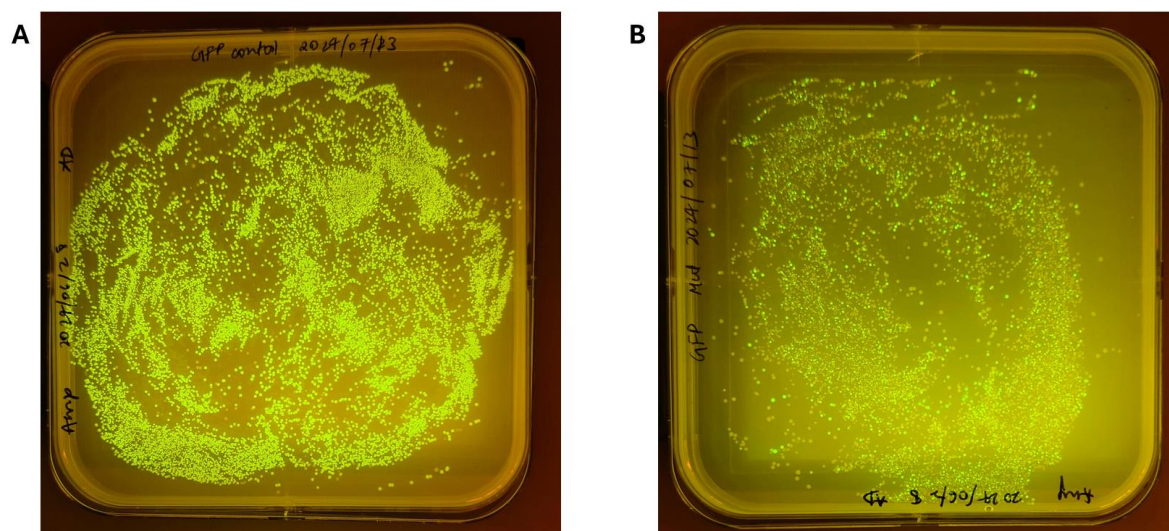


Fig 3- Transformed colonies under blue light. A) non-mutant control, B) Mutant library

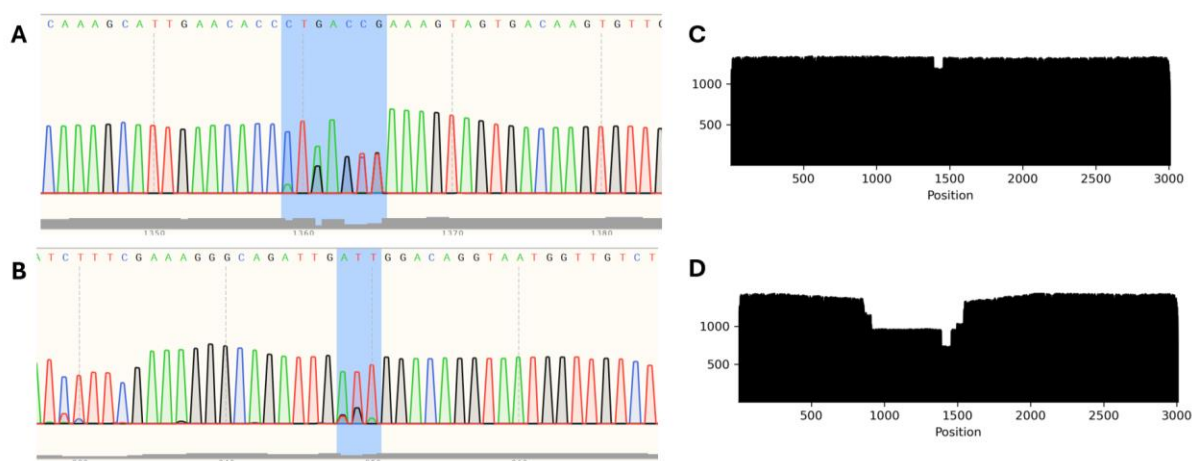


Fig 4- Whole plasmid Sequencing for mutant plasmid mixture. A) Chromophore region, B) 203rd amino acid region, C) sequence coverage plot for plasmid mixture from glowing cells, D) sequence coverage plot for plasmid mixture from non-glowing cells

Screening of GFP mutants

To determine if mutagenesis resulted in altered fluorescence, cells resuspended in M9 minimal media were excited at 405 nm, and their fluorescence was measured. A shift in the peak of the emission spectrum indicated the presence of different colored variants. Figure 5A shows the fluorescence emission spectra of 91 glowing colonies along with controls. The spectrum clearly demonstrates shifts in peak emission across different colonies. Figure 5B displays the emission spectra for controls: Dh5Alpha, GFP, and the cloned positive control. The cloned positive control exhibited a similar emission profile to GFP, with peak emission between 507 and 515 nm, indicating no shift. Variants were expected to show five distinct peak emissions: Blue, Cyan, Cyan-Green, Green, and Yellow. Figures 5C-H depict the emission spectra of various characterized mutants. Two colonies had a peak around 485 nm, indicating a Cyan fluorescence. Five colonies had peaks around 505 nm, corresponding to Cyan-Green fluorescence. Eight colonies showed peaks around 525 nm, which indicates yellow fluorescence. Forty-five colonies had peaks between 507 and 515 nm, corresponding to green fluorescence, while 31 colonies exhibited no clear peaks.

Colonies with no clear peaks appeared glowing under blue light and had significant fluorescent values, suggesting their peaks were outside the detected range. The blue peak was not captured with 405 nm excitation due to the excitation wavelength being too close to the emission peak, making differentiation difficult. To address this, excitation at 370 nm was employed (previously avoided due to its potential to damage cells). Figure 6 illustrates the excitation at both 370 nm and 405 nm for selected colonies. The characterized emission peaks were Blue at 440 nm, Cyan at 485 nm, Cyan-Green at 505 nm, Green at 513 nm, and Yellow at 527 nm. Among the selected mutants, one displayed a clear peak at 440 nm, indicating blue fluorescence. This peak was detectable with 370 nm excitation but not with 405 nm. Differentiation between Cyan-Green and Green fluorescence was more accurate with 370 nm excitation due to the closer proximity of their emission peaks. Colonies A12 and F9, previously characterized as Cyan-Green at 405 nm, exhibited emission curves more like Green when excited at 370 nm.

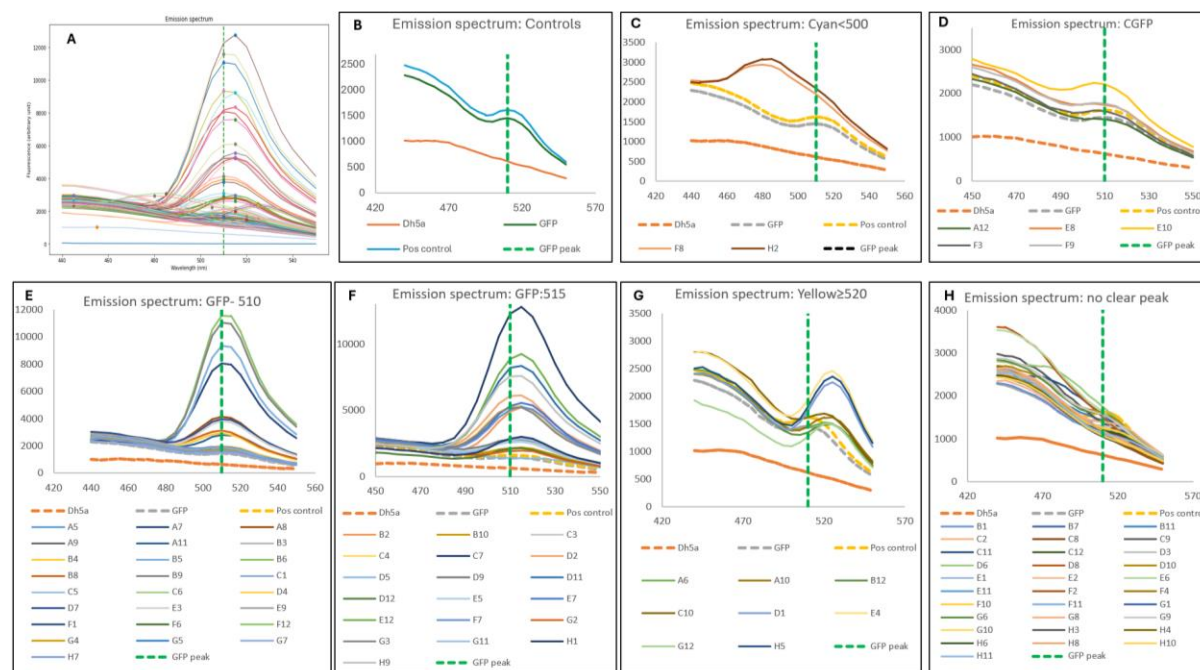


Fig 5- Emission spectrum analysis at 405nm excitation A) 96 well experiment, B) Controls, C) Cyan variants, D) Cyan-Green Variants, E) Green 510nm peak, F) Green: 515nm peak, G) Yellow Variants, H) colonies with no clear peak, I) Cyan variants, J) Cyan-Green Variants

Discussion

The study presented here explores the development of large protein libraries using the SLUPT (Synthesis of Libraries via deoxyuridine-containing PCR Templates) method, which offers a novel and efficient approach for generating multi-site mutations within DNA sequences. The primary objective was to establish a protocol capable of creating extensive and diverse libraries of mutant proteins with potential applications in protein engineering. Validation of this method was achieved through the creation of GFP mutants, with the goal of observing variations in fluorescence properties.

Protein engineering aims to enhance or introduce properties such as solubility, stability, and enzymatic activity. Directed evolution is a prevalent strategy to achieve these enhancements. Traditional library creation methods using random mutagenesis approaches often produce a high proportion of undesirable mutants. In contrast, targeted methods, which require prior knowledge of protein structure and function, provide high accuracy but can be complex and time-consuming when multiple regions need to be mutated. The SLUPT method addresses these challenges by generating single-stranded DNA (ssDNA) templates with deoxyuridine (dU) bases. These templates are then annealed with mutagenic primers, followed by gap filling and ligation to produce the desired mutant library. Unlike other methods, SLUPT does not rely on specific bacterial strains, viral systems, or restriction sites, making it broadly applicable. The process involves PCR amplification of the target DNA using dUTPs and phosphorylated primers, digestion of the 5' phosphorylated top strand with lambda exonuclease, and subsequent annealing with mutagenic primers. The resulting DNA is then extended, ligated, and digested with uracil-DNA glycosylase (UDG) to yield ssDNA containing the desired mutations. This method allows for the alteration of multiple DNA regions in a single reaction and ensures that all sequences within the library are present at equal concentrations, thereby enhancing screening efficiency.

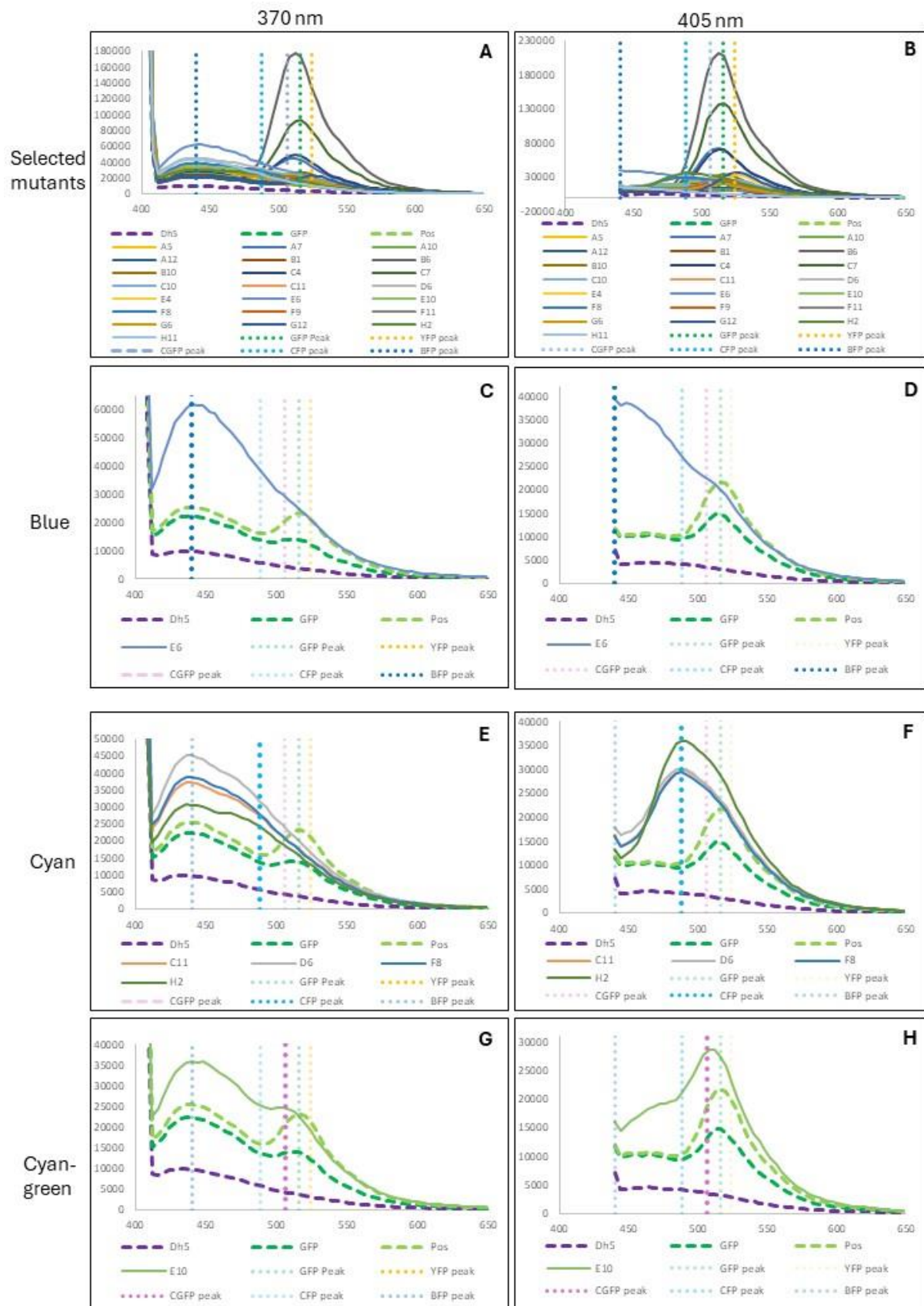


Fig 6- Emission spectrum analysis at 370 and 405nm excitation A) selected mutants at 370nm, B) selected mutants at 405nm, C) Blue variant at 370nm, D) Blue variant at 405nm, E) Cyan variant at 370nm, F) Cyan variant at 405nm, G) Cyan-green variant at 370nm, H) Cyan-green variant at 405nm

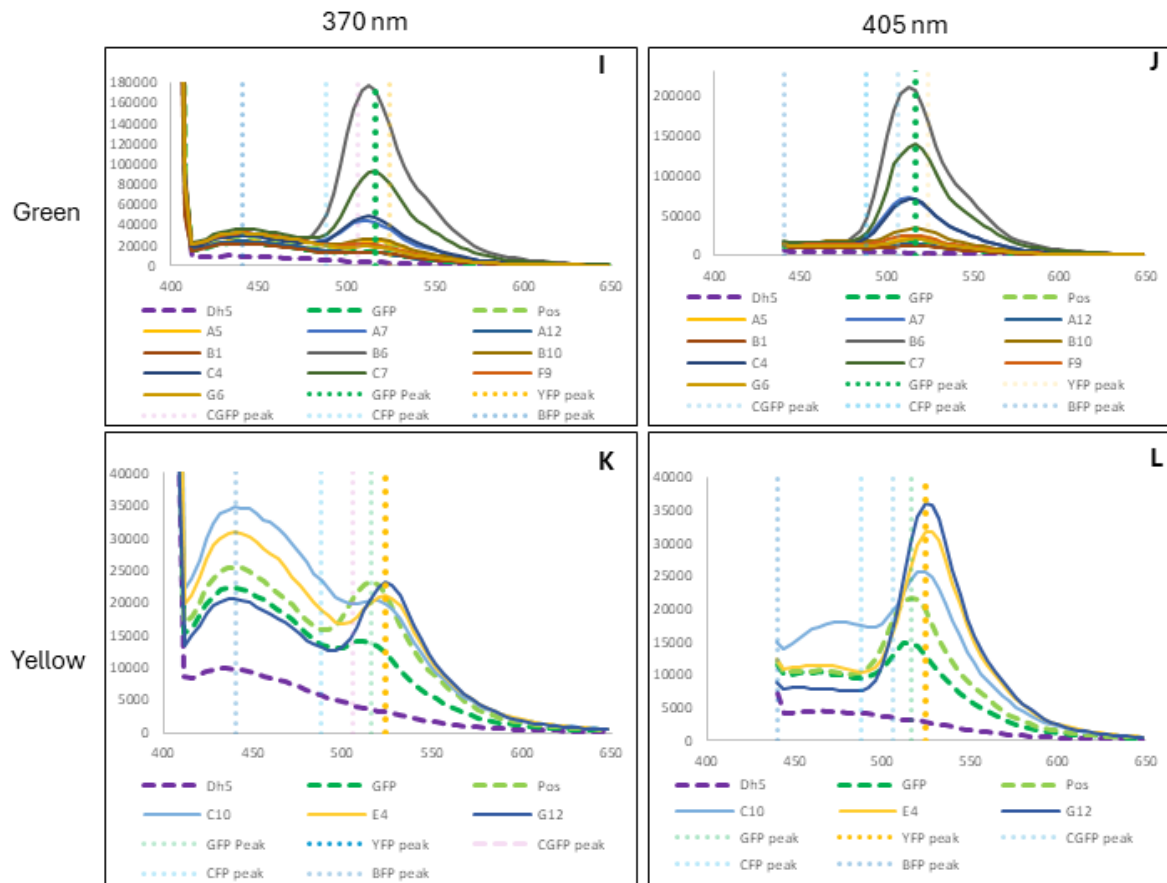


Fig 6 (continued)- Emission spectrum analysis at 370 and 405nm excitation I) Green variants at 370nm, J) Green variants at 405nm, K) Yellow variants at 370nm, L) Yellow variants at 405nm

We made two major modifications to this methodology. For generating the ssDNA when digesting with lambda exonuclease, instead of digesting it with 10 U enzyme for 1.5 hours (as mentioned in original protocol) we incubated it with 5 U enzyme for 30 mins as it was leading to over digestion of ssDNA, and it was degrading. The other modification we made was in PCR amplification when using ssDNA as template by performing first three cycles only with the complementary primer and then adding the other primer and resuming the PCR, this was done to ensure that a double stranded DNA template is formed which would be amplified and would increase the yield of the amplified sequence. The other modifications were made to add some control reactions which would help us figure out what and where things are going wrong (details mentioned in appendix). We applied this methodology to a new protein- GFP. After mutagenesis attempt, sequencing of plasmid mixtures from selected transformed colonies confirmed the presence of the desired mutations, with multiple peaks observed at the targeted sites, indicating effective mutagenesis. Interestingly, some colonies exhibited deletions in the GFP sequence, yet these deletions did not impair fluorescence, suggesting that truncated proteins may still retain functional properties. The cause of these deletions remains unclear, as they do not occur until after the insert has been digested; all tested samples were checked on an agarose gel and gel-extracted at the required size. The colonies displayed varying fluorescence, with emission spectra showing shifts in peak fluorescence, confirming the creation of GFP variants with different colors. Excitation at 370 nm allowed for capturing various peaks, including the blue peak, and better differentiation between closely spaced peaks such as cyan-green and green. Plasmids from some of these characterized mutants will be sent for sequencing to identify the mutations responsible for the changes in fluorescence and see whether they match with previously derived mutants. However, the

primary goal of this project was to assess the efficiency of the technique, rather than to characterize the specific mutations. This study demonstrates that the SLUPT method can effectively generate diverse GFP variants, showcasing its potential to create extensive protein libraries with altered functional properties.

The SLUPT method provides a robust and efficient strategy for generating large protein libraries with targeted mutations. By integrating both targeted and random mutagenesis, we can ensure comprehensive coverage of potential protein variants. Combining SLUPT with random mutagenesis can significantly expand the library size while achieving desired effects. SLUPT excels in targeted mutagenesis, and random mutagenesis can be seamlessly incorporated into this protocol without additional steps, enhancing library size and still producing relevant effects. This dual approach allows for a more thorough and unbiased exploration of protein variants, increasing the likelihood of discovering proteins with desirable traits. The technique holds considerable promise for applications in various fields of bioscience and bioengineering, including protein engineering, synthetic biology, and therapeutic development.

Appendix (Troubleshooting tricks)

1. Avoid over digestion when digesting with lambda exonuclease. Set up reactions with reduced concentration of enzyme and different incubation period if lower concentration of ssDNA is being observed.
2. When Annealing donor primers- Control reactions can be set to determine whether primers are annealing or not-
 - a. using one primer at a time (no forward primer): to ensure mutagenic primers are binding
 - b. and using primer for same region but without mutation (non-mutagenic primer): In case mutagenic primers do not bind, these reactions will help in determining if there are changes that need to be done in overlapping arms by changing the region.
3. PCR amplification at end stage: For control reactions without forward primers used in annealing should be amplified with 2 conditions:
 - a. donor primer + reverse primer: amplification suggest annealing is happening.
 - b. forward + reverse primer: no amplification is expected- UDG digestion is complete.
 - c. Troubleshooting:
 - i. Amplification in both cases: incubate longer for UDG digestion or increase enzyme amount.
 - ii. Amplification in none for mutants, but positive amplification in non-mutagenic primers: error in primer design, probably increase or change the length of overlapping complementary region.
 - iii. Amplification in none of the mutants or non-mutant control: try amplifying U containing ssDNA with complementary non-mutant donor primer-
 1. if positive, extension and ligation is not happening, add cofactors for enzyme and be careful with the storage of buffers
 2. if negative, then there is an issue with primer design.
4. Amplification using ssDNA- first few cycles of PCR using complementary primer only to facilitate formation of dsDNA, pause PCR and add the other primer and resume to amplify.
5. Cloning: first attempt at cloning- sequencing suggested missing part of insert- to be sure that things are not going wrong in between steps avoid PCR clean up, rather do gel extraction. Using gel extraction for insert increased the sequencing read with desired insert up to 3 times.

Reference

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